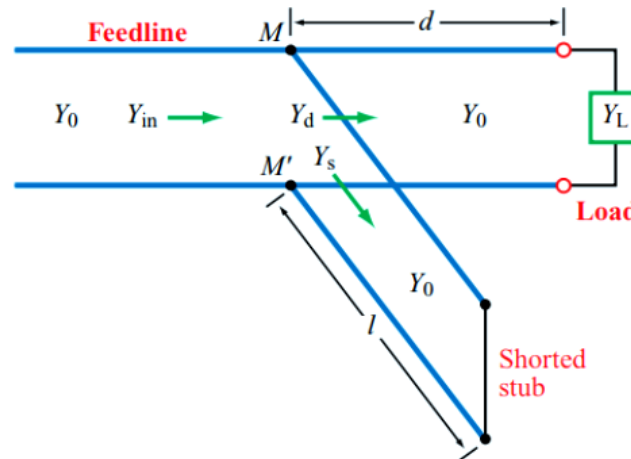


1. Design a single stub matching network for a load impedance $Z_L = 75 - j20 \Omega$ is connected to a $Z_o = 50 \Omega$ transmission line. Specify the short circuit stub insert location d (in wavelengths), length of stub l (in wavelengths), as well as all admittances. Use the Smith chart on the next page, annotate all your steps. Summarize your answers in the table below. All the transmission lines have characteristic have speed of travel $v_{ph} = 300$ meters/microsecond and frequency is 300 MHz.

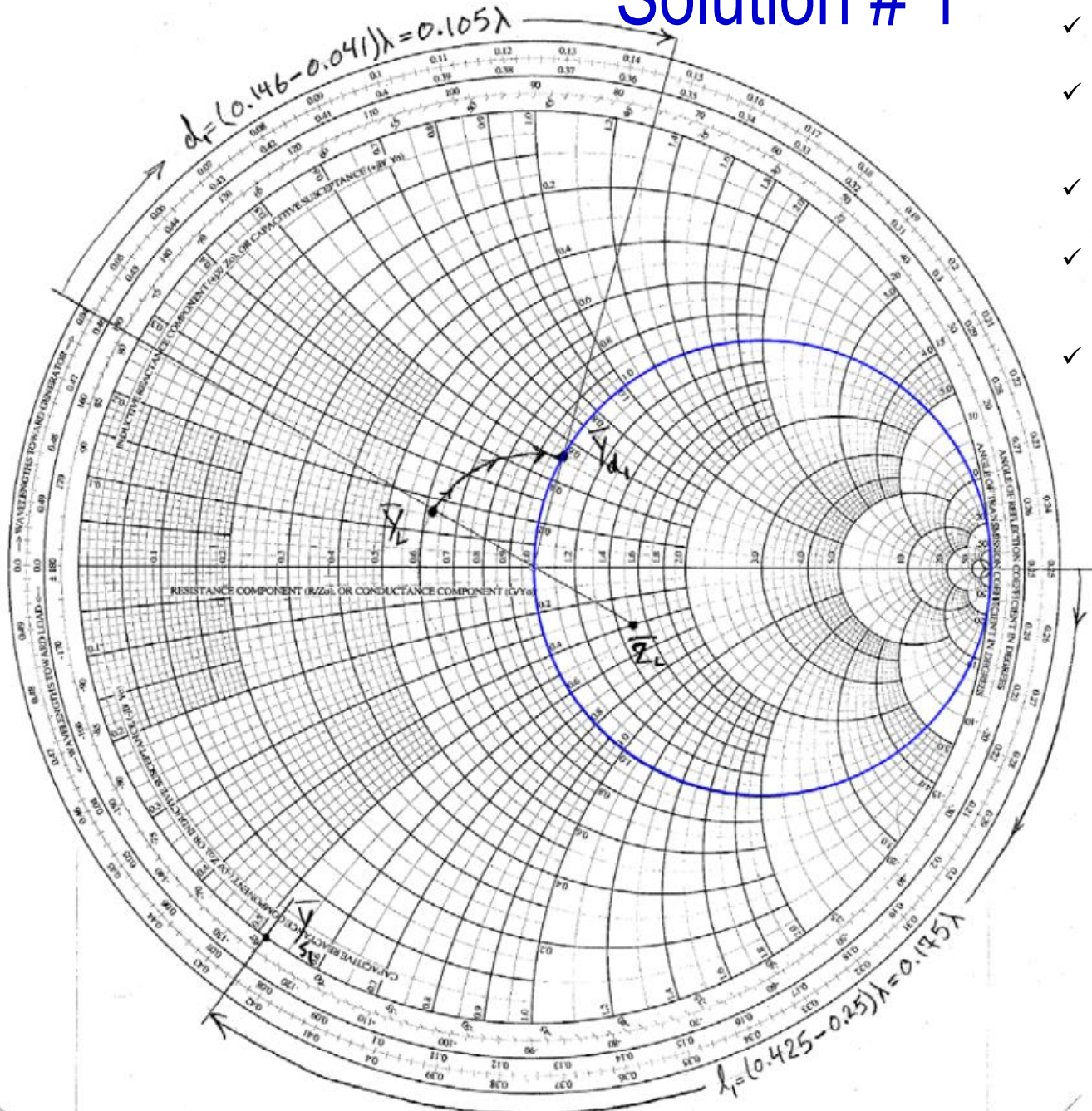


	d	y_d (normalized)	Y_d	y_s (normalized)	Y_s	l	y_{in} (normalized)	Y_{in}
Solution #1								
Solution #2								

1. Design a single stub matching network for a load impedance $Z_L = 75 - j20 \Omega$ is connected to a $Z_o = 50 \Omega$ transmission line. Specify the short circuit stub insert location d (in wavelengths), length of stub l (in wavelengths), as well as all admittances. Use the Smith chart on the next page, annotate all your steps. Summarize your answers in the table below. All the transmission lines have characteristic have speed of travel $v_{ph}=300$ meters/microsecond

1. Normalized load impedance $\bar{Z}_L = \frac{Z_L}{Z_o} = 1.5 - j0.4$
2. find the normalized admittance $\bar{y}_L$ on the Smith Chart (going through the center)
3. Draw the constant $|\Gamma|$ circle through point $\bar{y}_L$
4. Find the intersection of the constant $|\Gamma|$ circle with the $\bar{g}_L = 1$ circle
5. The first intersection is shown in the next slide. The point corresponds to $\bar{y}_{d_1}$
 - ✓ $\bar{y}_{d_1} = 1 + j0.525$ and is located at 0.146λ on WTG scale.
 - ✓ The distance between the load and the juncture is $d_1 = (0.146 - 0.042)\lambda = 0.105 \lambda$
 - ✓ We can find $\bar{y}_{s_1}$ which matches the feed line ($\bar{y}_{in} = 1$), $\bar{y}_{s_1} = -j0.525$
 - ✓ We can find the location of $\bar{y}_{s_1}$ on the smith chart which is located at 0.425λ on WTG scale
 - ✓ The length of the shorted stub is $l_1 = (0.425 - 0.25)\lambda = 0.175\lambda$

Solution # 1



- ✓ $\overline{y_{d_1}} = 1 + j0.525$ and is located at 0.146λ on WTG scale.
- ✓ The distance between the load and the juncture is $d_1 = (0.146 - 0.042)\lambda = 0.105 \lambda$
- ✓ We can find $\overline{y_{s_1}}$ which matches the feed line ($\overline{y_{in}} = 1$), $\overline{y_{s_1}} = -j0.525$
- ✓ We can find the location of $\overline{y_{s_1}}$ on the smith chart which is located at 0.425λ on WTG scale
- ✓ The length of the shorted stub is $l_1 = (0.425 - 0.25)\lambda = 0.175 \lambda$

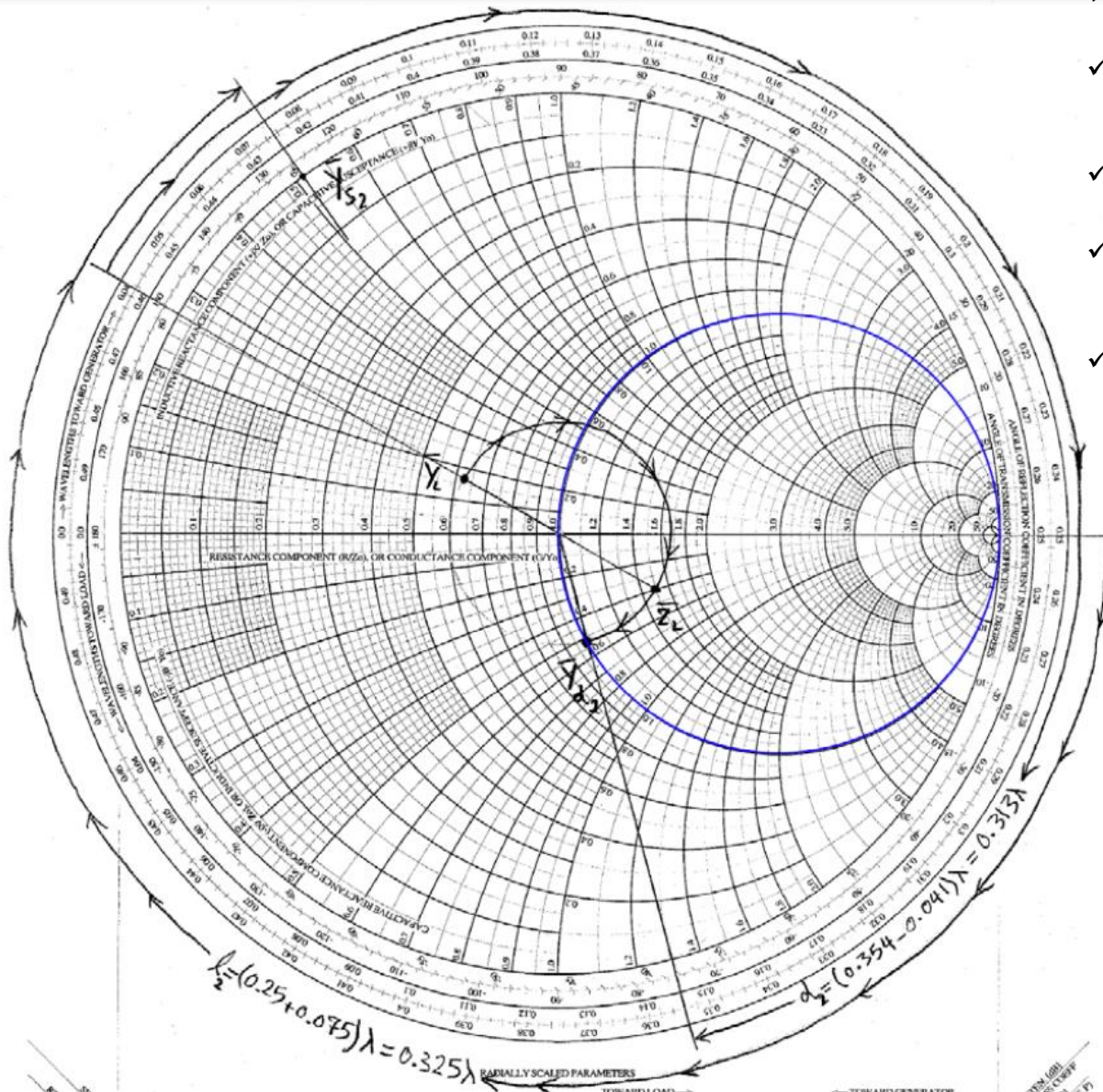
1. Design a single stub matching network for a load impedance $Z_L = 75 - j20 \Omega$ is connected to a $Z_o = 50 \Omega$ transmission line. Specify the short circuit stub insert location d (in wavelengths), length of stub l (in wavelengths), as well as all admittances. Use the Smith chart on the next page, annotate all your steps. Summarize your answers in the table below. All the transmission lines have characteristic have speed of travel $v_{ph}=300$ meters/microsecond

6. The second intersection is shown in the next slide. The point corresponds to

$\overline{y_{d_2}}$

- ✓ $\overline{y_{d_2}} = 1 - j0.525$ and is located at 0.354λ on WTG scale.
- ✓ The distance between the load and the juncture is $d_2 = (0.354 - 0.041)\lambda = 0.313\lambda$
- ✓ We can find $\overline{y_{s_2}}$ which matches the feed line ($\overline{y_{in}} = 1$), $\overline{y_{s_2}} = +j0.525$
- ✓ We can find the location of $\overline{y_{s_2}}$ on the smith chart which is located at 0.075λ on WTG scale
- ✓ The length of the shorted stub is $l_2 = (0.075 + 0.25)\lambda = 0.325\lambda$

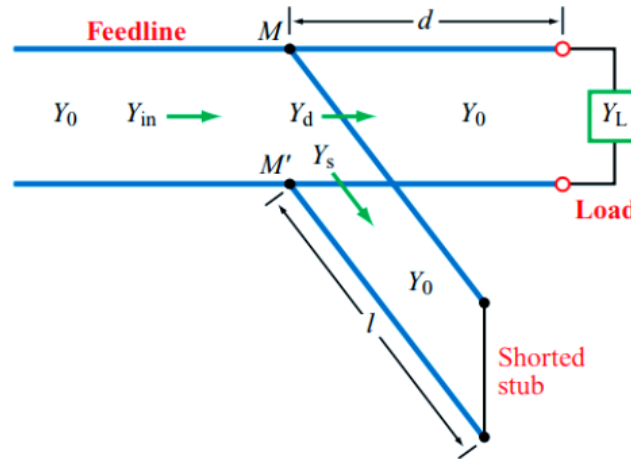
Solution # 2



- ✓ $\overline{y_{d_2}} = 1 - j0.525$ and is located at 0.354λ on WTG scale.
- ✓ The distance between the load and the juncture is $d_2 = (0.354 - 0.041)\lambda = 0.313\lambda$
- ✓ We can find $\overline{y_{s_2}}$ which matches the feed line ($\overline{y_{in}} = 1$), $\overline{y_{s_2}} = +j0.525$
- ✓ We can find the location of $\overline{y_{s_2}}$ on the smith chart which is located at 0.075λ on WTG scale
- ✓ The length of the shorted stub is $l_2 = (0.075 + 0.25)\lambda = 0.325\lambda$

1. Design a single stub matching network for a load impedance $Z_L = 75 - j20 \Omega$ is connected to a $Z_o = 50 \Omega$ transmission line. Specify the short circuit stub insert location d (in wavelengths), length of stub l (in wavelengths), as well as all admittances. Use the Smith chart on the next page, annotate all your steps. Summarize your answers in the table below. All the transmission lines have characteristic have speed of travel $v_{ph}=300$ meters/microsecond and frequency is 300 MHz.

✓ $\overline{y_{d1}} = 1 + j0.525$
 ✓ $d_1 = (0.146 - 0.042)\lambda = 0.105 \lambda$
 ✓ $\overline{y_{s1}} = -j0.525$
 ✓ $l_1 = (0.425 - 0.25)\lambda = 0.175\lambda$

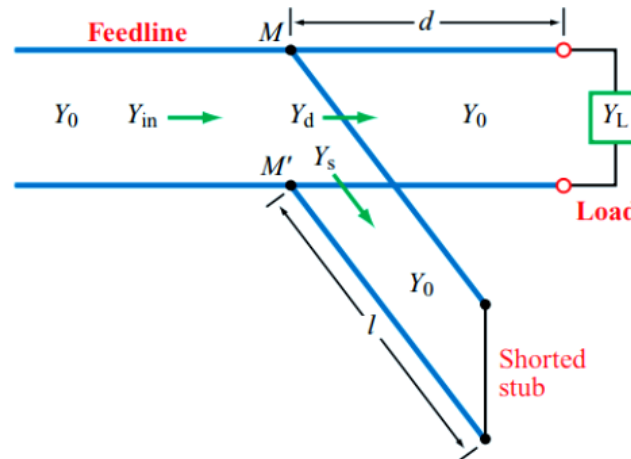


✓ $\overline{y_{d2}} = 1 - j0.525$
 ✓ $d_2 = (0.354 - 0.041)\lambda = 0.313\lambda$
 ✓ $\overline{y_{s2}} = +j0.525$
 ✓ $l_2 = (0.075 + 0.25)\lambda = 0.325\lambda$

$$\lambda = \frac{v_{ph}}{f} = \frac{300 * 10^6}{300 * 10^6} = 1 m$$

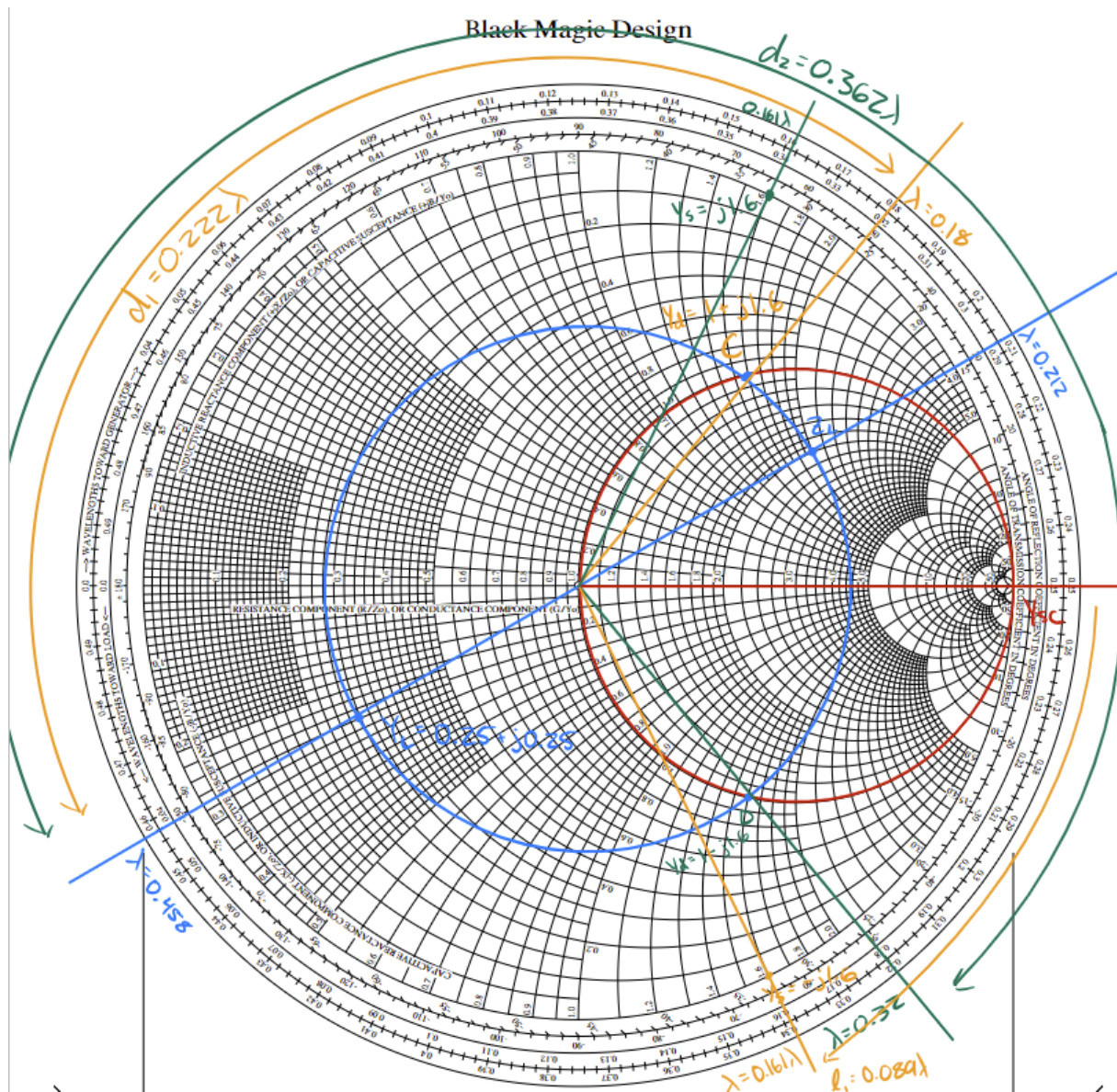
	d	y_d (normalized)	Y_d	y_s (normalized)	Y_s	l	y_{in} (normalized)	Y_{in}
Solution #1	0.105λ 10.5 cm	1 + j0.525	0.2 + j0.105 S	-j0.525	-j0.105 S	0.175λ 17.5 cm	1	0.02 S
Solution #2	0.313λ 31.3 cm	1 - j0.525	0.2 - j0.105 S	+j0.525	+j0.105 S	0.325λ 32.5cm	1	0.02 S

2. Design a single stub matching network for a load impedance $Z_L = 100 + j100 \Omega$ is connected to a $Z_o = 50 \Omega$ transmission line. Specify the short circuit stub insert location d (in wavelengths), length of stub l (in wavelengths), as well as all admittances. Use the Smith chart on the next page, annotate all your steps. Summarize your answers in the table below.

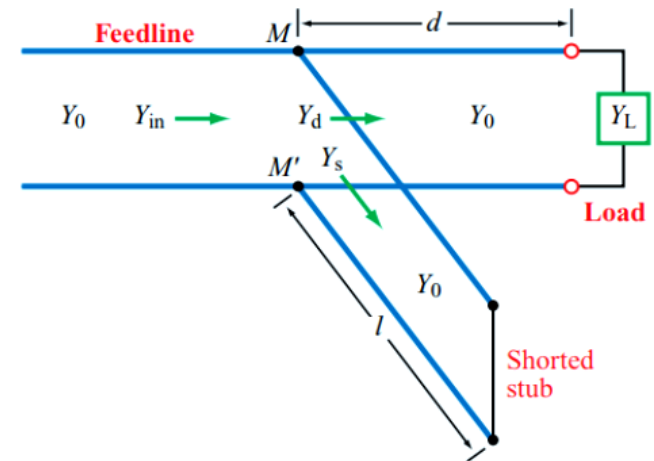
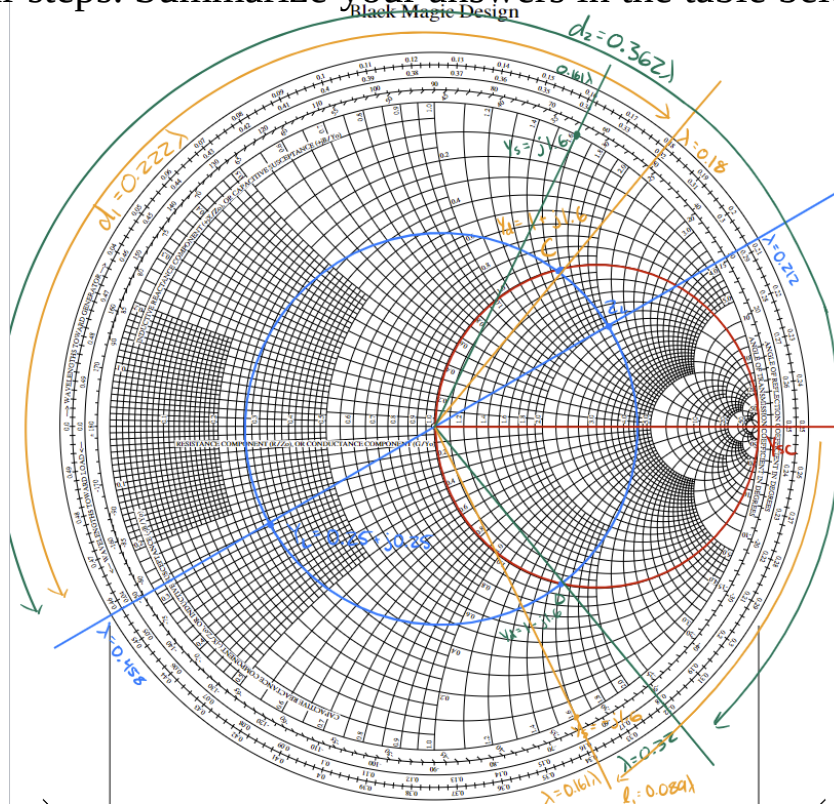


	d	y_d (normalized)	Y_d	y_s (normalized)	Y_s	l	y_{in} (normalized)	Y_{in}
Solution #1								
Solution #2								

Both Solutions

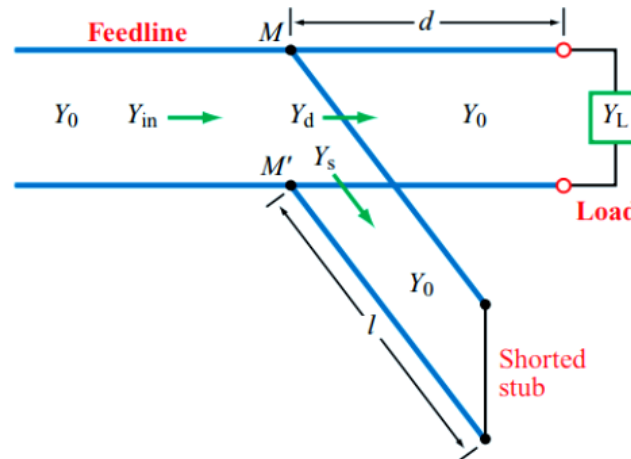


2. Design a single stub matching network for a load impedance $Z_L = 100 + j100 \Omega$ is connected to a $Z_0 = 50 \Omega$ transmission line. Specify the short circuit stub insert location d (in wavelengths), length of stub l (in wavelengths), as well as all admittances. Use the Smith chart on the next page, annotate all your steps. Summarize your answers in the table below.



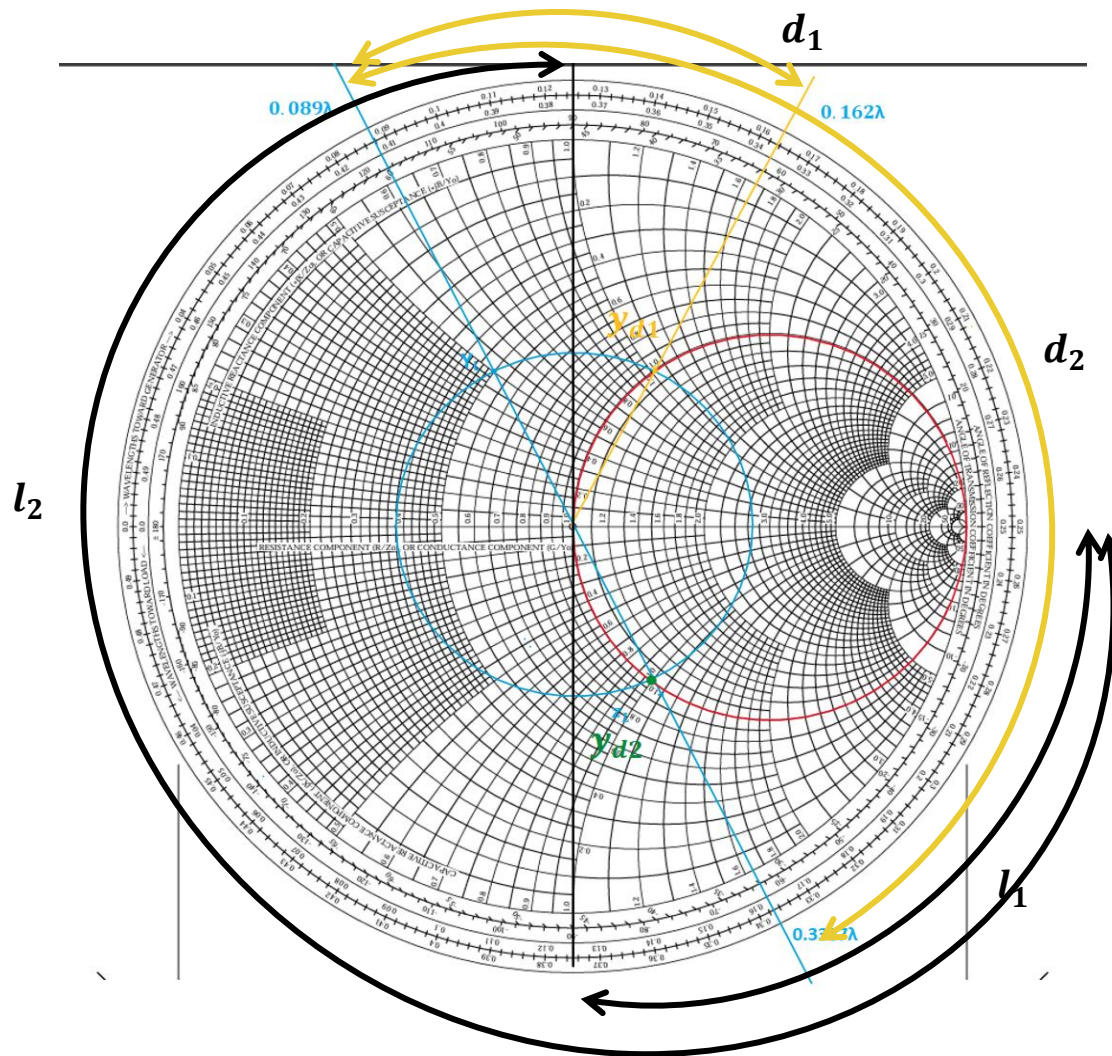
	d (in λ)	y_d (normalized)	Y_d	y_s (normalized)	Y_s	l (in λ)	y_{in} (normalized)	Y_{in}
Solution #1	0.222	$1 + j1.6$	$0.02 + j0.032$	$-j1.6$	$-j0.032$	0.089	1	0.02
Solution #2	0.362	$1 - j1.6$	$0.02 + j0.032$	$j1.6$	$j0.032$	0.411	1	0.02

3. Design a single stub matching network for a load impedance $Z_L = 50 - j50 \Omega$ is connected to a $Z_o = 50 \Omega$ transmission line. Specify the short circuit stub insert location d (in wavelengths), length of stub l (in wavelengths), as well as all admittances. Use the Smith chart on the next page, annotate all your steps. Summarize your answers in the table below.



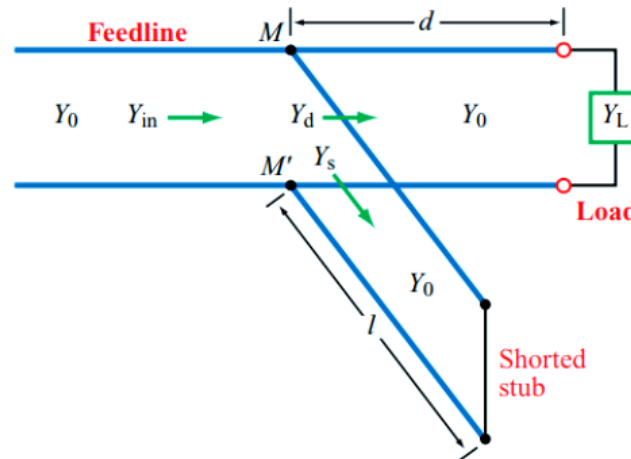
	d	y_d (normalized)	Y_d	y_s (normalized)	Y_s	l	y_{in} (normalized)	Y_{in}
Solution #1								
Solution #2								

Both Solutions



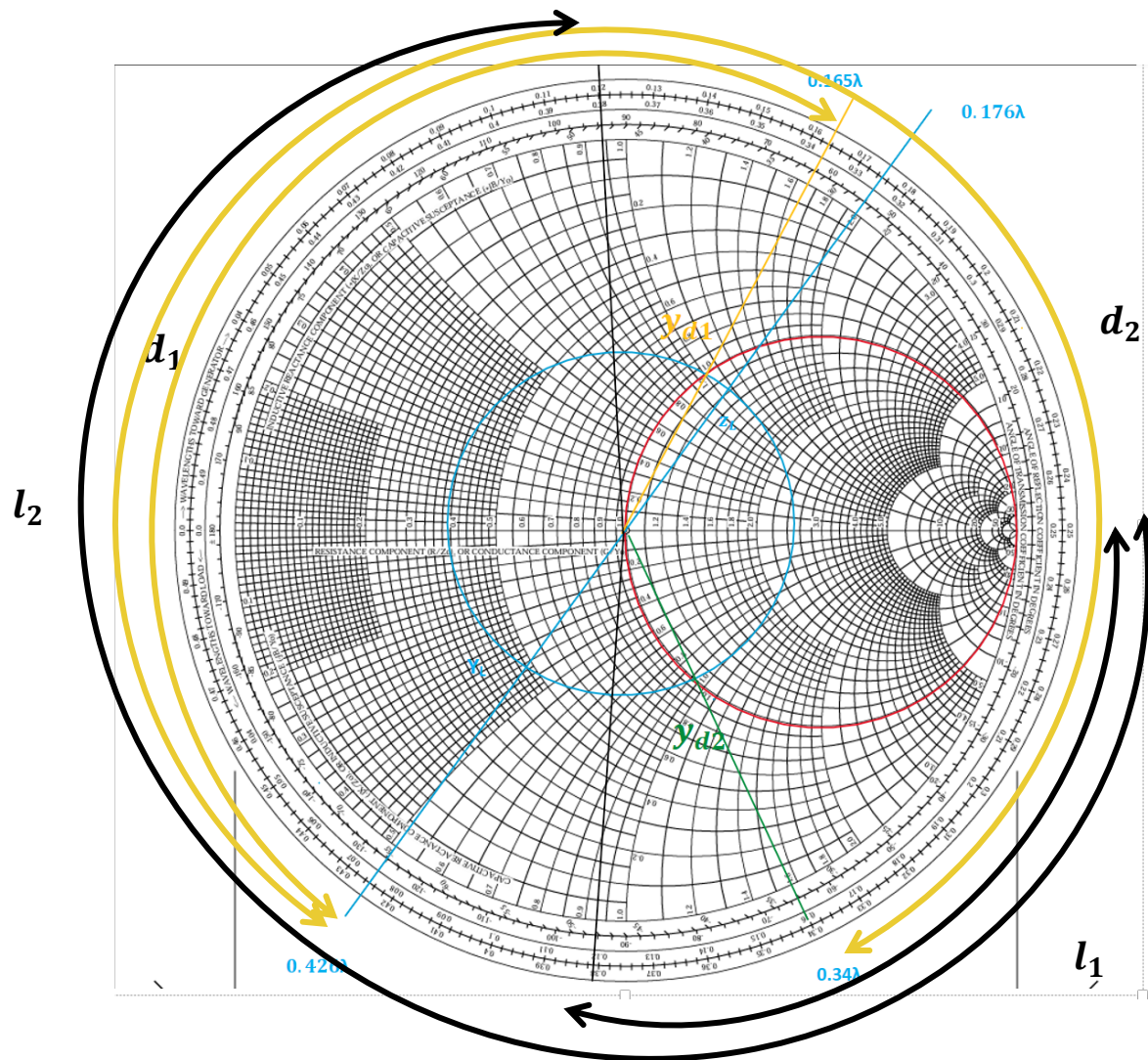
	d	y_d (normalized)	Y_d	y_s (normalized)	Y_s	l	y_{in} (normalized)	Y_{in}
Solution #1	0.073λ	$1+j$	$\frac{1}{50} + j\frac{1}{50}$	$-j$	$j\frac{1}{50}$	0.125λ	1	$\frac{1}{50}$
Solution #2	0.250λ	$1-j$	$\frac{1}{50} - j\frac{1}{50}$	j	$-j\frac{1}{50}$	0.375λ	1	$\frac{1}{50}$

4. Design a single stub matching network for a load impedance $Z_L = 60 + j50 \Omega$ is connected to a $Z_o = 50 \Omega$ transmission line. Specify the short circuit stub insert location d (in wavelengths), length of stub l (in wavelengths), as well as all admittances. Use the Smith chart on the next page, annotate all your steps. Summarize your answers in the table below.



	d	y_d (normalized)	Y_d	y_s (normalized)	Y_s	l	y_{in} (normalized)	Y_{in}
Solution #1								
Solution #2								

Both Solutions



	d	y_d (normalized)	Y_d	y_s (normalized)	Y_s	l	y_{in} (normalized)	Y_{in}
Solution #1	0.235λ	$1+0.93j$	$0.0186 j + \frac{1}{50}$	$-0.93j$	$0.0186 j$	0.131λ	1	$\frac{1}{50}$
Solution #2	0.416λ	$1-0.93j$	$0.0186 j - \frac{1}{50}$	$0.93j$	$- 0.0186 j$	0.369λ	1	$\frac{1}{50}$